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1. Let f(x) = x^2 sin(x). Find the derivative f’(x).

2. Evaluate the integral of x * exp(x) dx.

3. Prove that sqrt(2) is irrational.

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4. Solve dy/dx = y * (1 - y), y(0) = 2.

5. Find the limit of sin(3x) / x as x -> 0.

6. Compute the determinant of [[1,2],[3,4]].